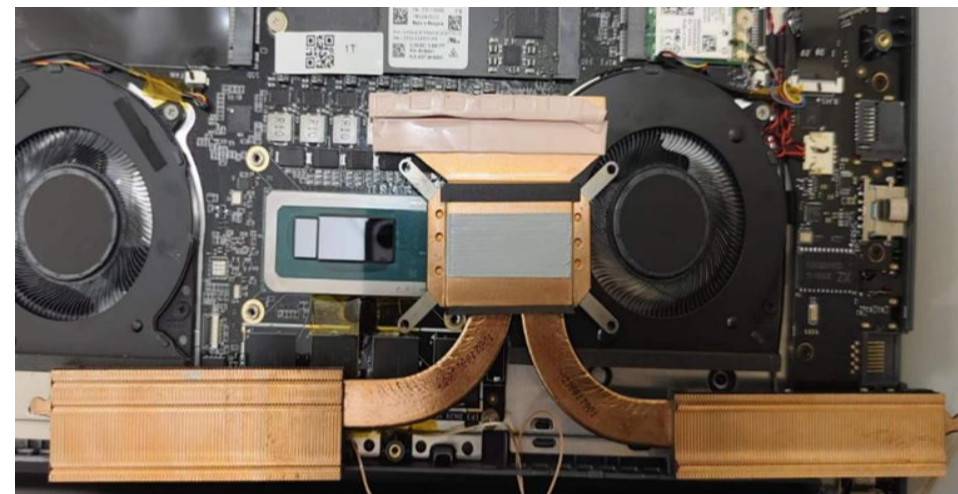
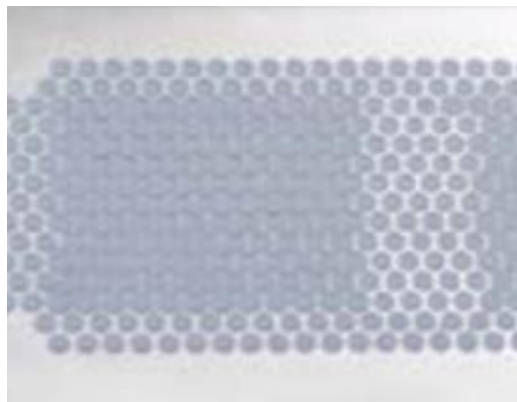
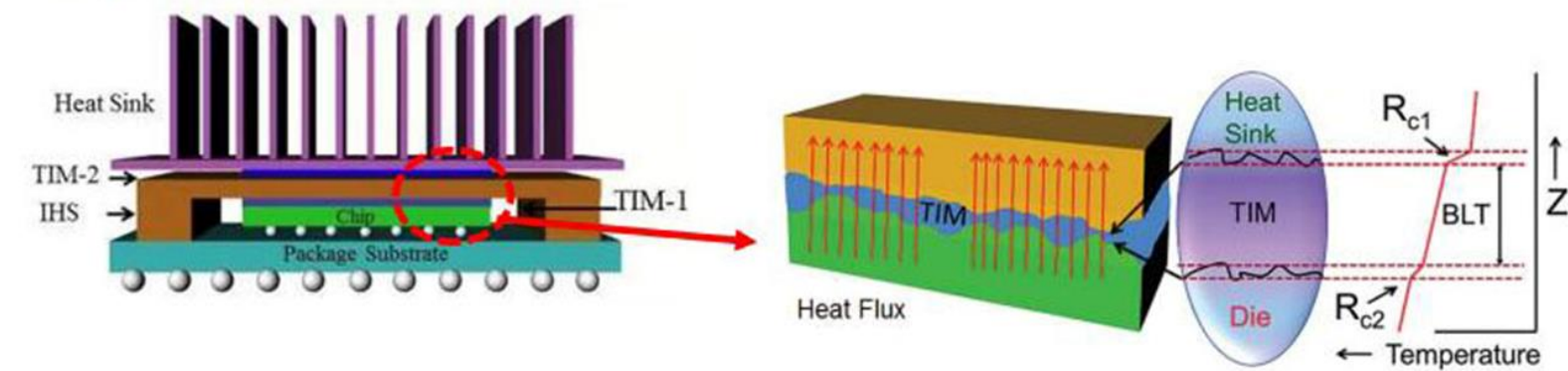


Deep-Materials TCG-LO Series Low Oil Bleeding Gels



Use fundamental science to make practical improvements in TIM

Deep-Materials

TIM Product Brochure

Use *fundamental science* to make practical improvements in Thermal Materials

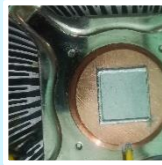
CPU/GPU TIMs



Grease

- Best in class performance
- Significantly reduce pump out

Highest performance traditional grease & Innovative Liquid Metal grease with low resistance



Liquid Metal

- Multi-phase Liquid metal – high viscosity
- Practical to use
- Highest performance

Brand new category of TIM with perf. of LM but ease of grease

Gap Fillers



Gels

- Highest conductivity
- Almost no oil seepage

Many kinds of unique gels – e.g. ultra low oil seepage gel, low dielectric constant gels



Gap Pads

- Almost eliminate oil seepage
- Reliable to very high temperatures

Ultra low oil seepage & ultra low outgassing silicone pads

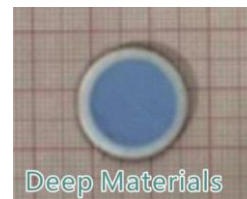
GAP Fillers

- For Applications in lower power/thicker TIM applications
- Do not require high pressure or mechanical support
- Typically used in consumer electronics, SSDs, Radios
- Performance measured in thermal conductivity (W/mK)

Our offerings in this space

- Liquid Gap Fillers/Gels
 - Single part (Fully cured)/Two part (in situ cure)
- Gap pads

TCG-LO Series: Low Oil Bleeding Gels



Applications:
Consumer Electronics, Automotive, Computing
SSDs, Memory Modules, Networking, Data Centers

	TCG-Gel-4000LO	TCG-Gel-4000LOVE	TCG-Gel-6000LO	Testing Method
Color	Green	Red	Blue	Visual
Minimum BLT (mm)	0.06	0.06	0.15	—
Density (g/cc)	3.25	3.25	3.3	ASTM D792
Flow Rate(g/min)	20	15	20	ASTM D2240
Operation Temp (°C)	-50~200	-50~200	-50~200	—
Electrical Performance				
Dielectric Constant@1GHZ	7.5	7.5	8.0	ASTM D150
Volume Resistance (Ω.cm)	10 ¹³	10 ¹³	10 ¹³	ASTM D257
Breakdown Voltage (KV/mm)	>8	>8	>8	ASTM D149
Flammability rating	V-0	V-0	V-0	UL 94
Thermal Performance				
Thermal Conductivity (W/m-K)	4.0	4.0	6.0	ASTM D5470
Oil Bleeding Size (mm)	< 1 mm	< 1 mm	< 2 mm	100°C aging for 24 h
Volatility (%)	< 0.08	< 0.05	< 0.08	ASTM E595

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